

Varun Pulipati

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SUMMARY

AI Data Solutions Engineer with 5 years of experience building applied AI/ML workflows, agent-enabled automation, enterprise data pipelines, and analytics products across loyalty, healthcare, fraud, and research domains. Hands-on with Python, SQL, PySpark, Databricks, Azure AI Foundry, FastAPI, MLflow, and Power BI. Strong at turning ambiguous business problems into governed, production-ready solutions through source-to-target mapping, data contracts, validation rules, model evaluation, UAT, technical documentation, and stakeholder enablement.

CORE STRENGTHS

AI/ML Engineering: predictive modeling, feature engineering, **Data Engineering:** ETL/ELT, ingestion pipelines, PySpark, Delta model validation, GenAI workflows, agentic automation Lake, data quality, schema contracts, lineage

Product Delivery: requirements, user stories, acceptance criteria, **Stakeholder Enablement:** executive reporting, demos, training UAT, technical documentation, roadmap execution walkthroughs, issue triage, cross-functional delivery

EXPERIENCE

AI & Data Solutions Engineer | RaceTrac - Atlanta, GA

Jul 2024 - Present

- **Built and supported enterprise data ingestion and transformation pipelines** across Punchh, Capillary, Braze, LoginRadius, POS, customer, transaction, offer, redemption, subscription, and engagement datasets using SQL, PySpark, Databricks, Delta Lake, Azure ADLS, and ADF.
- **Designed source-to-target mappings, data contracts, canonical schemas, KPI definitions, and validation rules** for a 5M+ member loyalty modernization program, improving data consistency across vendor, canonical, and reporting layers.
- **Developed applied ML and predictive analytics workflows** for churn propensity, retention targeting, cohort behavior, campaign measurement, anomaly detection, and forecast tracking using Python, SQL, feature engineering, and holdout logic.
- **Contributed to a 6-7% retention lift** by building a churn prediction framework with gradient boosted models, lift-by-decile evaluation, top-15% at-risk prioritization, and Braze-ready activation criteria.
- **Built an Azure AI Foundry agent workflow** that converts meeting transcripts, project notes, and discovery artifacts into requirements, solution summaries, user stories, development tasks, QA scenarios, and structured outputs with human review checkpoints.
- **Enabled natural-language analytics and self-service BI** by consolidating 60+ raw loyalty datasets into 20 curated Delta tables and supporting a Databricks AI/BI Genie layer for semantic metric discovery, business querying, and executive reporting.
- **Owned technical product and business analysis execution** by translating ambiguous stakeholder needs into requirements, process flows, acceptance criteria, backlog items, release-readiness checks, UAT plans, and production validation across engineering, QA, and business teams.
- **Led root-cause analysis and remediation planning** for production data issues including missing points, unrecognized transactions, offer qualification gaps, source-system discrepancies, and data-quality defects impacting customer-facing loyalty experiences.

Research Scientist | University of Missouri (NSF) - Columbia, MO

Aug 2022 - May 2024

- **Built an end-to-end analytics platform** that unified AR, motion-capture, Unity, Python, Airflow, and Tableau workflows to transform experimental sensor data into model-ready features, fatigue scores, and research dashboards.
- **Achieved approximately 80% precision** identifying high-fatigue states by engineering time-series motion features, training regression-based ML models, validating outputs against experimental labels, and publishing results across Ergonomics, HCII, and HFES.
- **Automated ingestion, cleansing, feature generation, model scoring, and reporting** using self-hosted Airflow DAGs, reducing manual research processing by roughly 70% across 4 concurrent studies.

Associate Data Scientist | Accenture (Aetna/CVS Health) - India

Aug 2020 - Dec 2021

- **Built XGBoost-based fraud risk models** on 2M+ healthcare claims using Python, SQL, provider billing anomalies, temporal claim patterns, feature engineering, model threshold tuning, and operations feedback loops.
- **Delivered SQL and Python data pipelines and Power BI dashboards** for claims ingestion, QA validation, investigation prioritization, false-positive tracking, model performance reporting, and leadership decision-making.
- **Partnered with fraud operations, compliance, and engineering teams** to convert investigator logic into analytics requirements, automate daily case queue delivery, and reduce pipeline turnaround from 5 days to under 24 hours.

PROJECTS

Unloop - AI Recommendation Engine | Chrome Extension - FastAPI - Behavioral AI 2025 - Present

- **Built an AI recommendation system** that converts browsing behavior into interaction-weighted weekly feature vectors, measures preference velocity, and surfaces explainable next-step recommendations through an API-backed product experience.
- **Designed the product architecture** across Chrome extension tracking, popup and overlay experiences, FastAPI services, Google OAuth, JWT authentication, behavioral events, ranking logic, and user-facing explanations.

Speak2py - LLM Analytics Engine | Open Source - PyPI - GenAI Jun 2025

- **Designed an LLM-powered analytics engine** that converts plain-English data questions into executable Pandas and SQL using schema-aware prompting, sandboxed execution, structured output validation, and automatic code checks.
- **Created an evaluation harness** comparing offline and cloud LLM outputs across accuracy, latency, error handling, and cost, helping determine when local models are sufficient versus cloud models.

ApplyPilot - Agentic Workflow Automation Platform | FastAPI - Playwright - Gemini May 2025

- **Built an AI workflow automation platform** that discovers ATS postings, tailors resumes and structured responses, executes browser workflows with Playwright, logs submissions to SQLite/Excel, and applies human-in-the-loop review checkpoints.
- **Designed controlled automation flows** by separating AI generation, user review, browser execution, logging, and retry steps so outputs could be improved without sacrificing quality or traceability.

EDUCATION & AWARDS

M.S. Computer Science, University of Missouri, Columbia USA (May 2024)

B.Tech Electronic & Computer Engineering, CVR College of Engineering India (May 2020)

- **Impact Award (2025)** recognized by Enterprise Director and business partners for loyalty migration delivery excellence
- Best Student Paper Award (HFES, 2024).

PUBLICATIONS

- **Publications (8)**, including 3 first-author papers across Ergonomics, HCII, and HFES.
- **Pulipati, V. et al.** Measuring Physical Demand in Augmented Reality Learning Environments. *Ergonomics*, 68(12), 2024.
- **Pulipati, V. et al.** Fatigue Dynamics of Augmented Reality Using Motion Capture. *Proceedings of HCI International*, 2024.
- **Pulipati, V. et al.** Physical Workload Assessment in AR Environments. *Proceedings of the Human Factors and Ergonomics Society*, 2024.

SELECTED AI & DATA WORKSTREAMS

- **Loyalty data modernization:** supported Punchh-to-Capillary migration, Braze CDI segmentation, LoginRadius identity mapping, POS transaction logic, subscription history, and customer event data across source, canonical, and reporting layers.
- **Predictive ML and activation:** developed churn, retention, campaign measurement, forecasting, and anomaly analysis workflows that connected Python/SQL outputs to Braze audiences, executive dashboards, and business decisioning.
- **AI-assisted SDLC automation:** shaped agent workflows that convert discovery inputs into business requirements, technical summaries, development tasks, QA cases, and release artifacts with review checkpoints.
- **Data quality and production validation:** built validation logic for transaction matching, offer qualification, points adjustment, missing event analysis, schema alignment, and issue remediation before downstream business impact scaled.

TECHNICAL SKILLS

AI Engineering: Azure AI Foundry, GenAI systems, agent workflows, RAG concepts, prompt engineering, structured output validation, human-in-the-loop review, LLM evaluation, recommendation systems, API-backed AI products

ML & Modeling: Python, Scikit-learn, XGBoost, TensorFlow, PyTorch, feature engineering, propensity modeling, classification, regression, model validation, A/B testing, causal inference, time-series forecasting, anomaly detection

Data Engineering: SQL, PySpark, Databricks, Spark, Delta Lake, Airflow, ETL/ELT pipelines, data quality validation, source-to-target mapping, data contracts, canonical data models, Azure ADLS/ADF, AWS S3/Redshift, Snowflake

Application, Reporting & Delivery: FastAPI, REST APIs, Docker, Git, MLflow, Streamlit, Power BI, Tableau, Databricks AI/BI Genie, KPI dashboards, executive reporting, recurring business reviews, data storytelling, requirements, user stories, acceptance criteria, UAT, Agile/Scrum, Azure DevOps, stakeholder communication